Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Solution Overview

The Emotion Detector is an AI-Driven Emotion Detection & Personalized Learning Support Platform. It bridges the gap between how students feel during learning and the kind of support they receive by using cutting-edge Natural Language Processing and Generative AI.

Core Components

Component 1: Dual-Model Emotion Classifier

A weighted ensemble of two independently trained deep learning models:

BiLSTM (Bidirectional Long Short-Term Memory)

- Architecture: Embedding(10000, 128) SpatialDropout(0.3) BiLSTM(128) 2 BatchNorm Dense(5, softmax)
- Input: Integer-encoded, padded token sequence (maxlen=100)
- Strength: Captures sequential and bidirectional context in text
- Training: EarlyStopping, ReduceLROnPlateau, class-weighting for balance

BERT (bert-base-uncased)

- Architecture: Pretrained transformer with fine-tuned classification head
- Input: Subword tokenized text with attention masks
- Strength: Deep contextual understanding of language semantics
- Training: HuggingFace Trainer API, FP16 mixed precision

Ensemble Strategy: 40% BiLSTM + 60% BERT weighted probability average

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Component 2: Rule-Based Keyword Enhancement

A dictionary of emotion-specific keywords for each of the 5 classes:

- Bored:** "boring", "dull", "uninteresting", "monotonous", "tedious"
- Confident:** "understand", "got it", "clear", "mastered", "ready"
- Confused:** "don't understand", "lost", "unclear", "confused", "makes no sense"
- Curious:** "wonder", "interesting", "want to know", "fascinating", "how does"

When keywords are matched, the corresponding emotion's probability is boosted by +15% and renormalized.

Component 3: Mixed Emotion Detection

If the probability gap between the top-2 detected emotions is < 15%, the system flags a "mixed emotion" state and reports both emotions to provide more nuanced guidance.

Component 4: Gemini-Powered Guidance Generator

Uses Google Gemini API (gemini-pro) to generate a structured 5-section response:

- Emotional Acknowledgement** - Validates the student's feeling
- Targeted Resources** - Specific study materials for their topic
- Study Techniques** - Emotion-matched learning strategies
- Break & Recovery Suggestions** - Mindful breaks if frustrated/bored
- Words of Encouragement** - Motivational closing

Component 5: Streamlit Web Application

A fully responsive, dark glassmorphism UI with:

- Home Page:** Input Detection Guidance in one flow
- Analytics Page:** 5 Plotly charts - emotion distribution, confidence trends, session history
- Model Comparison Page:** BiLSTM vs BERT architecture and performance comparison
- About Page:** Project description and usage guide

Component 6: Interaction Logger & Analytics

Every interaction is logged to data/interactions/interaction_logs.csv with fields:

timestamp, session_id, user_text, final_emotion, final_confidence, is_mixed, bilstm_emotion, bert_emotion, models_agree, rule_applied, guidance_excerpt

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Solution Differentiators

Feature	Generic Chatbot	Google Search	Emotion Detector
Emotion Detection			
Personalized to emotional state			
Dual AI models			
Rule-based enhancement			
Mixed emotion awareness			
Analytics dashboard			

Feature	Generic Chatbot	Google Search	Emotion Detector
No login required			
Free to use			